

SparkS7

SparkS7 is a Siemens S7-1200/1500 symbolic driver that browses PLC symbols by name, including optimized (non-absolute) blocks, without a TIA export.

Version 0.3.0 Ignition 8.1 and 8.3 Runs on: any gateway that reaches the PLC network

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1. Overview

SparkS7 is a standalone device driver, licensed and running independently of any other SparkBridge module. Module id `com.greenguardusa.sparks7`.

2. What You Need

- Ignition 8.1 and 8.3.
- An S7-1200/1500 PLC reachable over classic S7comm (ISO-on-TCP, TCP port 102).

3. Install

Install the signed module through Config > Modules > Install or Upgrade a Module; no build step is required. Module id `com.greenguardusa.sparks7`.

4. License

Licensed per gateway, bought once and kept, soft-licensed (`com.greenguardusa.sparkbridge.license`, entitlement `com.greenguardusa.sparks7`). Without a verified key it runs on Ignition's two-hour module trial; a present but invalid key fails closed and blocks the device until fixed or removed.

From release 2.7.0, every SparkBridge module runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). A verified key overrides the trial entirely and the trial clock is then ignored. Drop the key at `<data>/sparkbridge-license.key`; each module re-checks the file every 30 seconds, so installing, replacing, or removing a key takes effect with no restart.

5. Configure

Config > OPC UA > Device Connections > Create new Device > the S7 device type.

Setting Meaning, default

Host, Port PLC address; ISO-on-TCP default port 102.

Rack, Slot CPU endpoint selection; this is protocol routing, not authentication.

Security notes. Classic S7comm is plaintext with no credentials, encryption, or replay protection; this is a limitation of the Siemens protocol, not an omitted driver option. Isolate TCP 102 on a dedicated VLAN or VPN and firewall it so only the Ignition gateway can reach the PLC. Do not expose the PLC on an untrusted or internet-routable network.

S7comm+ (wire mode auto or plus) runs over TLS 1.3 tunneled in the S7 connection. The CPU certificate is trust-on-first-use: the driver logs its SHA-256 fingerprint on every connect until you copy it into the device's Certificate Fingerprint setting. Once pinned, a certificate that does not match fails the connection closed, and wire mode auto does not fall back to classic S7comm on a pin mismatch. Clear the setting to re-pin after a deliberate certificate rotation. S7comm+ is on by default since SparkS7 0.2.0; `sparks7.allowPlus=false` (or `SPARKS7_ALLOW_PLUS=false`) is the fleet-wide switch to turn it off.

6. Operate

Pin the CPU certificate fingerprint on every S7comm+ device once it is commissioned. A present but invalid license key fails the device closed rather than falling back to trial behavior, so remove or fix a bad key rather than leaving it in place.

7. Verify

See the README in the bundle for the verification procedure.

8. Troubleshoot

Symptom	Cause / fix
Device blocked with a valid-looking key present	An invalid key fails closed by design; remove or replace the key file rather than expecting trial fallback.
Device blocked with "certificate fingerprint ... does not match the pinned fingerprint"	The CPU presented a different certificate than the one pinned. If the device certificate was rotated on purpose, clear the Certificate Fingerprint setting and let it re-pin; otherwise treat it as a possible man-in-the-middle and check the network path before reconnecting.
Wire mode auto connects classic instead of S7comm+	The CPU refused S7comm+ (S7-300/400, or an S7-1500 with the S7comm+ server disabled). Background reconnects then stay classic until the device is restarted, which probes S7comm+ again.
Cannot reach the PLC from another VLAN	Expected if TCP 102 is isolated per the security notes; add an explicit, narrow firewall rule for the Ignition gateway only.

9. Reference

Module id `com.greenguardusa.sparks7`

Default port TCP 102 (ISO-on-TCP)

S7comm+ kill switch sparks7.allowPlus=false OR SPARKS7_ALLOW_PLUS=false (on by default since 0.2.0)

License entitlement com.greenguardusa.sparks7

10. Support

Docs page section: </sparkbridge/docs#sparks7>. Contact: </sparkbridge/contact>. Release downloads and checksums: [GitHub releases](#).